SPMeT Fast-Charging Simulator: Model Equations, Numerical Implementation, Parameters, and Reference Governor

Technical documentation for
https://github.com/scott-moura/SPMe-Charging-Dashboard

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Abstract

This document specifies the model and the controllers implemented in the browser-based fast-charging simulator (`spme.js`): a single particle model with electrolyte and lumped thermal dynamics (SPMeT), two side reactions that tally capacity loss (SEI growth and lithium plating), a constant-current/constant-voltage (CC-CV) charger, and a reference governor that enforces constraints on internal electrochemical states. It gives the continuous-time equations, the discretisation exactly as coded, the two parameter sets (an NMC811/graphite-SiO_x cell and a $LiFePO_4$/graphite cell) with their sources, and benchmark numbers against which a re-implementation can be checked.

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1 Scope and conventions

- **Sign convention.** The applied current I [A] is *positive for charging*. This is the opposite of the convention in most of the DFN literature and in PyBaMM.
- **Geometry.** The through-thickness coordinate x runs from the anode current collector ($x = 0$) through the anode ($0 < x < L_n$), the separator ($L_n < x < L_n + L_s$) and the cathode ($L_n + L_s < x < L$), with $L = L_n + L_s + L_p$. Subscripts n, s, p denote anode, separator and cathode; $j \in \{n, p\}$ denotes an electrode.
- **Units.** SI throughout: concentrations in $\mathrm{mol\,m^{-3}}$, potentials in V, temperature in K. Capacity-loss tallies are in Ah.
- **Constants.** $F = 96485.33212\,\mathrm{C\,mol^{-1}}$, $R_g = 8.314462618\,\mathrm{J\,mol^{-1}K^{-1}}$, $T_{\mathrm{ref}} = 298.15\,\mathrm{K}$.

States. The model state is

$$x = \left[\, c_s^n(r,t),\; c_s^p(r,t),\; c_e(x,t),\; T(t),\; Q_{\mathrm{sei}}(t),\; Q_{\mathrm{pl}}(t) \,\right], \tag{1}$$

that is, the lithium concentration in one representative spherical particle per electrode, the electrolyte concentration across the cell, the lumped cell temperature, and two capacity-loss tallies. After discretisation (Section 3) the state has $2N_r + N_e + 3 = 66$ entries.

Derived geometric quantities. For each electrode, the specific interfacial area and the total active surface are

$$a_j = \frac{3\,\varepsilon_{s,j}}{R_j}, \qquad S_j = a_j L_j A, \tag{2}$$

where $\varepsilon_{s,j}$ is the active-material volume fraction, R_j the particle radius, L_j the electrode thickness and A the electrode plate area.

2 Model equations

The model follows the SPMe structure of Moura et al. [1] and Marquis et al. [2], obtained from the Doyle–Fuller–Newman (DFN) model [3] by assuming that the interfacial reaction current is uniform across each electrode.

2.1 Solid-phase diffusion

In each electrode, lithium diffuses in a sphere of radius R_j:

$$\frac{\partial c_s^j}{\partial t} = \frac{D_{s,j}(T)}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial c_s^j}{\partial r}\right), \qquad 0 < r < R_j, \tag{3}$$

$$\left.\frac{\partial c_s^j}{\partial r}\right|_{r=0} = 0, \qquad D_{s,j}(T)\left.\frac{\partial c_s^j}{\partial r}\right|_{r=R_j} = J_j, \tag{4}$$

where J_j is the molar flux *into* the particle,

$$J_n = \frac{I}{F\,S_n}, \qquad J_p = -\frac{I}{F\,S_p}. \tag{5}$$

The surface concentration is $c_{ss}^j = c_s^j(R_j, t)$, the volume average is $\bar{c}_s^j = \frac{3}{R_j^3}\int_0^{R_j} r^2 c_s^j \,\mathrm{d}r$, and the corresponding stoichiometries are $\theta_{ss,j} = c_{ss}^j / c_{\max,j}$ and $\bar{\theta}_j = \bar{c}_s^j / c_{\max,j}$.

2.2 Electrolyte diffusion

$$\varepsilon_e(x)\,\frac{\partial c_e}{\partial t} = \frac{\partial}{\partial x}\left(D_e^{\mathrm{eff}}\,\frac{\partial c_e}{\partial x}\right) + s(x)\,I, \qquad D_e^{\mathrm{eff}} = D_e(c_e)\,\psi_{D_e}(T)\,\varepsilon_e(x)^b, \tag{6}$$

with zero flux at both current collectors, $\partial c_e/\partial x = 0$ at $x = 0$ and $x = L$, and continuity of concentration and flux at the electrode | separator interfaces. The source term per ampere is uniform in each electrode,

$$s(x) = \begin{cases} -\dfrac{1 - t_+}{F A L_n}, & 0 < x < L_n \quad \text{(anode)},\\[2mm] 0, & L_n < x < L_n + L_s \quad \text{(separator)},\\[2mm] +\dfrac{1 - t_+}{F A L_p}, & L_n + L_s < x < L \quad \text{(cathode)}. \end{cases} \tag{7}$$

Here ε_e is the electrolyte volume fraction, b the Bruggeman exponent and t_+ the cation transference number. Because $\int_0^L s\,\mathrm{d}x = 0$, the electrolyte lithium inventory $A\int_0^L \varepsilon_e c_e\,\mathrm{d}x$ is conserved.

2.3 Temperature dependence

All temperature-dependent parameters follow an Arrhenius law,

$$\psi_E(T) = \exp\left[\frac{E}{R_g}\left(\frac{1}{T_{\mathrm{ref}}} - \frac{1}{T}\right)\right], \tag{8}$$

applied to $D_{s,j}$, the reaction rate constants k_j, D_e, the electrolyte conductivity κ and the SEI rate, each with its own activation energy E (Section 5).

2.4 Interfacial kinetics

Butler–Volmer kinetics with symmetric transfer coefficients ($\alpha = 0.5$) and a uniform reaction current give a closed form for the overpotential:

$$i_{0,j} = k_j\,\psi_{k_j}(T)\,\sqrt{\bar{c}_{e,j}}\,\sqrt{c^j_{ss}\,(c_{\max,j} - c^j_{ss})}, \tag{9}$$

$$\eta_n = -\frac{2R_gT}{F}\,\sinh^{-1}\!\left(\frac{I}{2\,S_n\,i_{0,n}}\right), \qquad \eta_p = +\frac{2R_gT}{F}\,\sinh^{-1}\!\left(\frac{I}{2\,S_p\,i_{0,p}}\right), \tag{10}$$

where $\bar{c}_{e,j}$ is the electrolyte concentration averaged over electrode j. In (9) and in the open-circuit potentials, the surface stoichiometry is clipped to $[10^{-4},\,1-10^{-4}]$. The interfacial potential differences ($\phi_s - \phi_e$, averaged over each electrode) are

$$\Phi_n = U_n(\theta_{ss,n}) + \eta_n, \qquad \Phi_p = U_p(\theta_{ss,p}) + \eta_p, \tag{11}$$

with U_j the open-circuit potentials given in Section 5.

2.5 Electrolyte potential

The electrolyte potential satisfies the concentrated-solution form of Ohm's law [3] with unit thermodynamic factor,

$$\frac{\partial \phi_e}{\partial x} = \frac{I}{A}\,\frac{f(x)}{\kappa^{\text{eff}}} + \frac{2R_gT}{F}\,(1-t_+)\,\frac{\partial \ln c_e}{\partial x}, \qquad \kappa^{\text{eff}} = \kappa(c_e)\,\psi_\kappa(T)\,\varepsilon_e(x)^b, \tag{12}$$

where $f(x) \in [0,1]$ is the fraction of the applied current carried by the electrolyte. With a uniform reaction current it is piecewise linear,

$$f(x) = \min\!\left(1,\ \frac{x}{L_n},\ \frac{L-x}{L_p}\right). \tag{13}$$

(During charging the ionic current flows in the $-x$ direction, $i_e = -(I/A)f(x)$, which gives the sign of the first term.) The potential is referenced arbitrarily; only differences enter the outputs. Unlike the closed-form approximations in [1, 2], (12) is integrated numerically on the electrolyte grid with concentration-dependent κ (Section 3.3).

2.6 Terminal voltage

With $\bar{\phi}_{e,j}$ the average of ϕ_e over electrode j,

$$V = \Phi_p - \Phi_n + (\bar{\phi}_{e,p} - \bar{\phi}_{e,n}) + I\,R_s, \qquad R_s = \frac{1}{A}\left(\frac{L_n}{3\,\sigma_n^{\text{eff}}} + \frac{L_p}{3\,\sigma_p^{\text{eff}}}\right), \tag{14}$$

where $\sigma_j^{\text{eff}} = \sigma_j\,(1-\varepsilon_{e,j})^{b_s}$ is the effective electronic conductivity and the $L/3\sigma$ form is the electrode-averaged solid-phase ohmic drop of [2].

2.7 Plating overpotential

Lithium plating is thermodynamically possible where $\phi_s - \phi_e < 0$ against Li/Li$^+$ [9]. It is most negative at the anode | separator interface. Taking the solid potential as uniform across the anode, the model evaluates

$$\eta_{\text{pl}} = \Phi_n - (\phi_e(L_n) - \bar{\phi}_{e,n}). \tag{15}$$

This construction is specific to this tool; it is not taken from [1] or [2]. Its accuracy against a DFN model is discussed in Section 6.

2.8 Lumped thermal model

$$C_{\mathrm{th}}\frac{\mathrm{d}T}{\mathrm{d}t} = \underbrace{I\,(V - U_{\mathrm{ocv}})}_{\dot{Q}_{\mathrm{gen}}} - hA_s\,(T - T_{\mathrm{amb}}), \qquad U_{\mathrm{ocv}} = U_p(\bar{\theta}_p) - U_n(\bar{\theta}_n). \tag{16}$$

This is the energy balance of Bernardi et al. [7] without the entropic (reversible) term. Because U_{ocv} is evaluated at the *bulk-average* stoichiometries, $\dot{Q}_{\mathrm{gen}}$ includes the heat released as concentration gradients inside the particles relax ("heat of mixing").

2.9 Side reactions and capacity loss

Two cathodic side reactions at the anode surface are evaluated per unit of anode active area and integrated into capacity-loss tallies. They are *not* fed back: lithium is not removed from the anode and no film resistance is added.

$$i_{\mathrm{sei}} = i_{0,\mathrm{sei}}\;\psi_{\mathrm{sei}}(T)\;\exp\!\left[-\frac{\alpha_{\mathrm{sei}}F}{R_gT}\,(\Phi_n - U_{\mathrm{sei}})\right], \tag{17}$$

$$i_{\mathrm{pl}} = \begin{cases} \min\left\{\, i_{0,\mathrm{pl}}\left[\exp\!\left(-\dfrac{\alpha_{\mathrm{pl}}F}{R_gT}\,\eta_{\mathrm{pl}}\right) - 1\right],\ \dfrac{\max(I,0)}{S_n}\,\right\}, & \eta_{\mathrm{pl}} < 0, \\ 0, & \eta_{\mathrm{pl}} \geq 0, \end{cases} \tag{18}$$

$$\frac{\mathrm{d}Q_{\mathrm{sei}}}{\mathrm{d}t} = \frac{i_{\mathrm{sei}}\,S_n}{3600}, \qquad \frac{\mathrm{d}Q_{\mathrm{pl}}}{\mathrm{d}t} = \frac{i_{\mathrm{pl}}\,S_n}{3600} \qquad [\mathrm{Ah\,s^{-1}}]. \tag{19}$$

Equation (17) is the reaction-limited (Tafel) SEI model of Ramadass et al. [8] in the form used in [6]. Equation (18) is an irreversible plating reaction in the spirit of [9, 10], with no stripping; the plating current is capped at the applied current density. The arguments of both exponentials are capped at 40 for numerical safety.

2.10 State of charge, capacity and initial condition

State of charge is defined on the anode,

$$\mathrm{SOC} = \frac{\bar{\theta}_n - \theta_{n,0}}{\theta_{n,100} - \theta_{n,0}}, \tag{20}$$

the cell capacity is that of the anode stoichiometry window, and the cathode stoichiometry at 0 % SOC follows from the lithium balance:

$$Q = \frac{\varepsilon_{s,n} L_n A\, c_{\max,n} F\,(\theta_{n,100} - \theta_{n,0})}{3600}\ [\mathrm{Ah}], \qquad \theta_{p,0} = \theta_{p,100} + \frac{3600\,Q}{\varepsilon_{s,p} L_p A\, c_{\max,p} F}. \tag{21}$$

A charge from SOC_0 starts at rest: both particles uniform at $\theta_j = \theta_{j,0} + \mathrm{SOC}_0(\theta_{j,100} - \theta_{j,0})$, $c_e = c_{e,0}$ everywhere, $T = T_{\mathrm{amb}}$, $Q_{\mathrm{sei}} = Q_{\mathrm{pl}} = 0$. C-rates are defined against Q from (21).

3 Numerical implementation

Both diffusion equations are discretised with conservative finite volumes and advanced with implicit (backward) Euler. Each sub-step requires one tridiagonal solve (Thomas algorithm). The sub-models are coupled by operator splitting within a time step (Section 3.4).

3.1 Particles: finite volumes in r

Each particle is divided into $N_r = 20$ concentric shells of equal thickness $\Delta r = R_j/N_r$. Shell $i \in \{0, \dots, N_r - 1\}$ occupies $i\Delta r < r < (i+1)\Delta r$ and carries the cell-average concentration c_i. Normalising shell volumes by Δr^3 and face areas by Δr^2 (the factor 4π cancels),

$$w_i = \frac{(i+1)^3 - i^3}{3}, \qquad \alpha_i = i^2 \quad (i = 0, \dots, N_r), \qquad \sum_i w_i = \frac{N_r^3}{3}. \tag{22}$$

With $\mu = D_{s,j}(T^k)\,\Delta t/\Delta r^2$, backward Euler on the flux-form of (3) gives, for each shell,

$$-\mu\alpha_i\, c_{i-1}^{k+1} + (w_i + \mu\alpha_i + \mu\alpha'_{i+1})c_i^{k+1} - \mu\alpha'_{i+1}\, c_{i+1}^{k+1} = w_i\, c_i^k + \delta_{i,N_r-1}\,\frac{\Delta t\, N_r^2}{\Delta r}\, J_j^k, \tag{23}$$

where $\alpha'_{i+1} = \alpha_{i+1}$ for $i < N_r - 1$ and $\alpha'_{N_r} = 0$ (the surface flux enters through the right-hand side), and $\alpha_0 = 0$ enforces symmetry at the centre. Summing (23) over i shows that $\sum_i w_i c_i$ changes by exactly $\Delta t\, N_r^2 J_j/\Delta r$ per step, i.e. $\Delta\bar{c}_s = 3J_j\Delta t/R_j$: the scheme conserves lithium to round-off for any Δt.

The volume average and the surface concentration are

$$\bar{c}_s = \frac{3}{N_r^3}\sum_{i=0}^{N_r-1} w_i c_i, \qquad c_{ss} = c_{N_r-1} + \frac{J_j\,\Delta r}{2\,D_{s,j}(T)}, \tag{24}$$

the latter a half-cell extrapolation from the outermost cell centre using the imposed surface flux.

3.2 Electrolyte: finite volumes in x

The electrolyte grid has $N_n = 10$, $N_s = 3$ and $N_p = 10$ uniform cells in the anode, separator and cathode ($N_e = 23$). Cell j has width Δx_j, volume fraction ε_j, Bruggeman factor ε_j^b and source s_j from (7). The diffusive conductance between cells j and $j+1$ is the series (harmonic) combination of two half cells,

$$g_{j+\frac{1}{2}} = \left(\frac{\Delta x_j}{2D_j^{\text{eff}}} + \frac{\Delta x_{j+1}}{2D_{j+1}^{\text{eff}}}\right)^{-1}, \qquad D_j^{\text{eff}} = D_e\big(\max(c_j^k, 1)\big)\,\psi_{D_e}(T^k)\,\varepsilon_j^b, \tag{25}$$

which handles the jump in porosity at the interfaces without special treatment. The diffusivity is lagged (evaluated at the old concentration and temperature), so the step is linear:

$$-g_{j-\frac{1}{2}}c_{j-1}^{k+1} + (m_j + g_{j-\frac{1}{2}} + g_{j+\frac{1}{2}})c_j^{k+1} - g_{j+\frac{1}{2}}c_{j+1}^{k+1} = m_j c_j^k + s_j\, I^k\,\Delta x_j, \qquad m_j = \frac{\varepsilon_j \Delta x_j}{\Delta t}, \tag{26}$$

with $g_{-\frac{1}{2}} = g_{N_e-\frac{1}{2}} = 0$. Since $\sum_j s_j \Delta x_j = 0$, the inventory $\sum_j \varepsilon_j \Delta x_j c_j$ is conserved exactly.

Conservative positivity limiter. At high rate or low temperature the electrolyte at the anode collector can deplete, and (26) can then return non-physical concentrations. After the solve, any cell below the floor $c_{\text{fl}} = 1\,\text{mol}\,\text{m}^{-3}$ is raised to the floor using lithium taken from its neighbour, first sweeping toward the cathode and then back:

$$\text{for } j = 0, \dots, N_e - 2: \ \text{if } c_j < c_{\text{fl}}: \quad c_{j+1} \leftarrow c_{j+1} - (c_{\text{fl}} - c_j)\frac{\varepsilon_j \Delta x_j}{\varepsilon_{j+1}\Delta x_{j+1}}, \quad c_j \leftarrow c_{\text{fl}}; \tag{27}$$

$$\text{for } j = N_e - 1, \dots, 1: \ \text{if } c_j < c_{\text{fl}}: \quad c_{j-1} \leftarrow c_{j-1} - (c_{\text{fl}} - c_j)\frac{\varepsilon_j \Delta x_j}{\varepsilon_{j-1}\Delta x_{j-1}}, \quad c_j \leftarrow c_{\text{fl}}. \tag{28}$$

This keeps $\ln c_e$ defined without creating lithium. (A plain clamp $c_j \leftarrow \max(c_j, c_{\text{fl}})$ adds several percent of lithium in depleting cases.) Once the electrolyte is depleted the uniform-reaction assumption has failed and results are indicative only.

3.3 Electrolyte potential: quadrature of (12)

Define the half-cell ohmic resistance and the diffusion-potential coefficient

$$\rho_j = \frac{\Delta x_j}{2\,\kappa_j^{\mathrm{eff}}}, \qquad \kappa_j^{\mathrm{eff}} = \max\!\big(\kappa(c_j)\,\psi_\kappa(T),\,10^{-4}\big)\,\varepsilon_j^{b}, \qquad k_d = \frac{2R_gT}{F}(1-t_+). \tag{29}$$

Cell-centre potentials are accumulated from $\phi_0 = 0$:

$$\phi_j = \phi_{j-1} + \frac{I}{A}\,f_{j-\frac{1}{2}}\,(\rho_{j-1}+\rho_j) + k_d\,(\ln c_j - \ln c_{j-1}), \qquad j = 1,\dots,N_e-1, \tag{30}$$

with $f_{j-\frac{1}{2}} = \min(1,\, x_{j-\frac{1}{2}}/L_n,\, (L - x_{j-\frac{1}{2}})/L_p)$ evaluated at the face between cells $j-1$ and j. The electrode averages are arithmetic means over the cells of each electrode, $\bar\phi_{e,n} = \frac{1}{N_n}\sum_{j<N_n}\phi_j$ and $\bar\phi_{e,p} = \frac{1}{N_p}\sum_{j\ge N_n+N_s}\phi_j$; likewise $\bar c_{e,n}$ and $\bar c_{e,p}$ in (9). The potential at the anode | separator face, needed in (15), is

$$\phi_e(L_n) = \phi_{N_n-1} + \frac{I}{A}\,\rho_{N_n-1} + \tfrac{1}{2}\,k_d\,(\ln c_{N_n} - \ln c_{N_n-1}), \tag{31}$$

i.e. the ohmic drop across the last anode half cell (where $f \approx 1$) plus half of the diffusion-potential step across the face.

3.4 Order of operations in one time step

Given the state x^k, a current I^k held constant over the step, the step size Δt and T_{amb}, the function `step()` performs:

1. **Particles.** Solve (23) for each electrode with $D_{s,j}(T^k)$ and J_j^k from (5).
2. **Electrolyte.** Solve (26) with T^k, then apply the limiter.
3. **Outputs.** Evaluate (24), (9)–(11), the potential quadrature, (14), (15), (17)–(18) and $\dot Q_{\mathrm{gen}}$ at the *new* concentrations, the current I^k and the *old* temperature T^k. These are the outputs reported for time t^{k+1}.
4. **Temperature.** Backward Euler on the cooling term of (16) with the heat generation from step 3:

$$T^{k+1} = \frac{T^k + \dfrac{\Delta t}{C_{\mathrm{th}}}\,(\dot Q_{\mathrm{gen}} + hA_sT_{\mathrm{amb}})}{1 + \dfrac{\Delta t}{C_{\mathrm{th}}}\,hA_s}. \tag{32}$$

5. **Capacity loss.** Forward Euler: $Q_{\mathrm{sei}}^{k+1} = Q_{\mathrm{sei}}^{k} + i_{\mathrm{sei}}S_n\Delta t/3600$, and likewise for Q_{pl}.

The scheme is first-order accurate in time. There is no nonlinear iteration: every sub-step is a linear solve or an explicit formula, which is what makes it cheap enough to re-run on every slider movement (about 10^3 steps per charge; a governor run costs roughly thirty model steps per time step because of the trial simulations of Section 4).

3.5 Time step

With a base step Δt_0, the step is Δt_0 for $t < 1800\,\mathrm{s}$, $2\Delta t_0$ for $1800 \le t < 5400\,\mathrm{s}$ and $4\Delta t_0$ afterwards. The two displayed charges use $\Delta t_0 = 2\,\mathrm{s}$; the CC-CV sweep for the trade-off chart and the search for the fastest in-limits CC-CV rate use $\Delta t_0 = 4\,\mathrm{s}$. Simulations stop after at most 5 h.

4 Charging controllers

4.1 Formulation

Both chargers are implemented by the same routine. At each time step k, with the full state x_k known, the applied current is the largest value in $[0, I_{\max}]$ for which the model predicts that every constrained output stays admissible over a look-ahead of H steps under that constant current:

$$I_k = \max\Big\{\, I \in [0, I_{\max}] \;:\; y(\hat{x}_{k+h|k}(I), I) \in \mathcal{Y}, \quad h = 1, \dots, H \Big\}, \tag{33}$$

where $\hat{x}_{k+h|k}(I)$ is obtained by applying the model step of Section 3.4 h times from x_k with current I, and y is the output map evaluated at the end of each step. Writing $I_k = \beta_k I_{\max}$ with $\beta_k \in [0, 1]$, this is the modified reference governor of Perez et al. [11]: the reference is the charger's maximum current $I_{\max}$ and β_k is the largest admissible scaling. See [12] for the general theory of reference governors.

Reference governor. The admissible set constrains internal states:

$$\mathcal{Y}_{\mathrm{RG}} = \Big\{\, \eta_{\mathrm{pl}} \geq \eta_{\min}, \quad T \leq T_{\max}, \quad c_e(0) \geq c_{e,\min}, \quad \Phi_p \leq \Phi_{p,\max}, \quad \theta_{ss,n} \leq \theta_{n,\max} \,\Big\}, \tag{34}$$

with look-ahead $H = 3$ steps. Here $c_e(0)$ is the first electrolyte cell (anode current collector), $\Phi_{p,\max} = U_p(\theta_{p,100})$ is the cathode's rest potential at 100 % SOC, $c_{e,\min} = 100\,\mathrm{mol\,m^{-3}}$ and $\theta_{n,\max} = 0.99$. The limits $\eta_{\min}$ and $T_{\max}$ are user inputs (defaults $+5\,\mathrm{mV}$ and $45\,^{\circ}\mathrm{C}$). There is *no terminal-voltage constraint*; the cathode-potential constraint plays that role.

CC-CV. The admissible set is $\mathcal{Y}_{\mathrm{CCCV}} = \{V \leq V_{\max}\}$ with $H = 1$ and $I_{\max}$ equal to the CC current. While the predicted voltage is below $V_{\max}$ this returns $I_{\max}$ (constant current); afterwards it returns the current that places the end-of-step voltage at $V_{\max}$ to within the bisection tolerance (constant voltage).

4.2 Algorithm

Problem (33) is solved by bisection on I, assuming that admissibility is monotone in the current (if I is admissible, so is every smaller current). The bracket is warm-started around the previous current. Algorithm 1 is the procedure exactly as implemented in `simulate()`.

Algorithm 1 Closed-loop charge with the scalar reference governor (and CC-CV)

Require: initial state x_0 (rest at SOC_0, $T = T_{\mathrm{amb}}$), target SOC_T, $I_{\max}$, horizon H, admissible set $\mathcal{Y}$

$I_{\mathrm{prev}} \leftarrow I_{\max}$; $t \leftarrow 0$; $t_{\mathrm{stall}} \leftarrow 0$
function TRIAL(I) ▷ forward simulation on a copy of the state
  $\tilde{x} \leftarrow x$
  for $h = 1, \ldots, H$ **do**
    $(\tilde{x}, y) \leftarrow \mathrm{STEP}(\tilde{x}, I, \Delta t, T_{\mathrm{amb}})$
    if $y \notin \mathcal{Y}$ **then**
      record first violated constraint; **return** false
    end if
  end for
  return true
end function
while $t < t_{\max}$ **do**
  choose Δt from the schedule for the current t
  if TRIAL($I_{\max}$) **then**
    $I \leftarrow I_{\max}$ ▷ no constraint active
  else
    $lo \leftarrow 0$; $hi \leftarrow I_{\max}$
    $a \leftarrow \max(0,\, I_{\mathrm{prev}} - 0.04\, I_{\max})$; $b \leftarrow \min(I_{\max},\, I_{\mathrm{prev}} + 0.04\, I_{\max})$ ▷ warm start
    if $a > 0$ **then if** TRIAL(a) **then** $lo \leftarrow a$ **else** $hi \leftarrow a$
    end if
    if $hi > b$ **and** $b > lo$ **then if** TRIAL(b) **then** $lo \leftarrow b$ **else** $hi \leftarrow b$
    end if
    $n \leftarrow 0$
    while $n < 12$ **and** $hi - lo > 2 \times 10^{-3}\, I_{\max}$ **do**
      $m \leftarrow (lo + hi)/2$; **if** TRIAL(m) **then** $lo \leftarrow m$ **else** $hi \leftarrow m$; $n \leftarrow n + 1$
    end while
    $I \leftarrow lo$; active constraint $\leftarrow$ the one violated by TRIAL(hi)
  end if
  $(x, y) \leftarrow \mathrm{STEP}(x, I, \Delta t, T_{\mathrm{amb}})$; $t \leftarrow t + \Delta t$; $I_{\mathrm{prev}} \leftarrow I$; record y
  if $\mathrm{SOC} \geq \mathrm{SOC}_T$ **then stop** (target reached)
  end if
  if $I < Q/20$ **then** $t_{\mathrm{stall}} \leftarrow t_{\mathrm{stall}} + \Delta t$ **else** $t_{\mathrm{stall}} \leftarrow 0$
  end if
  if $t_{\mathrm{stall}} \geq 120\,\mathrm{s}$ **then stop** (current collapsed below C/20)
  end if
end while

Implementation notes.

- The constraints in (34) are tested in the order written; the first one violated by the upper end of the final bracket is reported as the active constraint.
- $I = 0$ is assumed admissible and is never tested. If the constraints are already violated at rest (for example $T_{\max} \leq T_{\mathrm{amb}}$), the governor returns $I \approx 0$ and the charge stops through the C/20 rule.

- The resolution of the current is $2 \times 10^{-3} I_{\max}$. The governor therefore rides its active constraint from the admissible side with a small sawtooth; reported minima such as "$\min \eta_{\mathrm{pl}} = +5.0\,\mathrm{mV}$" hold to that resolution.
- The look-ahead is $H\Delta t$ (6 s at the base step, longer once the step coarsens). Constraints are checked only at step ends, and the prediction uses the same model and the same T_{amb} as the plant: there is no model mismatch, no state estimator, and no robustness margin other than the user's choice of $\eta_{\min}$ and $T_{\max}$. Recursive feasibility is not proven; in practice the short horizon suffices because the constrained outputs respond to the current almost instantaneously (η_{pl}, Φ_p through the kinetic and ohmic terms) or slowly and monotonically (T, c_e).
- When a constraint is active, a governor step costs one trial at $I_{\max}$, two warm-start trials and about six bisection trials (the warm-start bracket is $0.08\,I_{\max}$ wide), each of up to $H = 3$ model steps.

4.3 Derived comparisons shown in the tool

- **Capacity lost** is $Q_{\mathrm{sei}} + Q_{\mathrm{pl}}$ at the end of the charge; **time below zero** is the sum of the step sizes whose end-of-step $\eta_{\mathrm{pl}} < 0$.
- **Trade-off curve.** CC-CV charges at a fixed list of C-rates (NMC: 0.5 to 3C; LFP: 0.5 to 6C) with $\Delta t_0 = 4\,\mathrm{s}$.
- **Fastest in-limits CC-CV.** The largest CC rate $c \in [0.1, c_{\mathrm{hi}}]$ whose whole trajectory satisfies $\min_t \eta_{\mathrm{pl}} \geq \eta_{\min} - 0.1\,\mathrm{mV}$ and $\max_t T \leq T_{\max} + 0.02\,\mathrm{K}$, found by 7 bisection steps on c ($c_{\mathrm{hi}} = 3$ for NMC, 6 for LFP), with $\Delta t_0 = 4\,\mathrm{s}$.

5 Parameter values

Two cells are implemented. Table 1 lists the electrode parameters, Table 2 the cell-level, thermal and side-reaction parameters, and Table 3 the discretisation and controller settings. Source tags:

C Chen et al. 2020 [4] (LG M50), checked against PyBaMM's `Chen2020.py` [13].

O O'Kane et al. 2022 [6] (same cell), checked against PyBaMM's `OKane2022.py`.

P PyBaMM's `Prada2013.py` (PyBaMM 26.10), which attributes its values to Prada et al. 2013 [14] and borrows the graphite open-circuit potential and kinetics from [4], the $LiFePO_4$ open-circuit potential from [16] and the $LiFePO_4$ kinetics from [17]. The full text of [14] was not available when this set was assembled, so these values rest on PyBaMM's transcription.

P12 Prada et al. 2012 [15] (same A123 cell), Tables III and V.

A Assumption made for this tool.

D Derived from the other values via (21).

5.1 Open-circuit potentials

Graphite-SiO_x anode (both cells), from [4]:

$$U_n(\theta) = 1.9793\, e^{-39.3631\,\theta} + 0.2482 - 0.0909 \tanh(29.8538(\theta - 0.1234)) - 0.04478 \tanh(14.9159(\theta - 0.2769)) - 0.0205 \tanh(30.4444(\theta - 0.6103)). \quad (35)$$

Table 1: Electrode and separator parameters (reference temperature 298.15 K).

		NMC cell (LG M50)			LFP cell (A123 26650)		
Parameter	Unit	Anode	Sep.	Cathode	Anode	Sep.	Cathode
Thickness L	μm	85.2	12	75.6	34	25	80
Electrolyte fraction ε_e	–	0.25	0.47	0.335	0.36	0.45	0.426
Active fraction ε_s	–	0.75	–	0.665	0.58	–	0.374
Particle radius R	μm	5.86	–	5.22	5	–	0.05
$c_{\max}$	$\mathrm{mol\,m^{-3}}$	33133	–	63104	30555	–	22806
D_s	$\mathrm{m^2\,s^{-1}}$	3.3×10^{-14}	–	4.0×10^{-15}	3.0×10^{-15}	–	5.9×10^{-18}
E of D_s	$\mathrm{kJ\,mol^{-1}}$	30.3	–	25	35	–	39
k	(see note)	6.48×10^{-7}	–	3.42×10^{-6}	6.48×10^{-7}	–	6×10^{-7}
E of k	$\mathrm{kJ\,mol^{-1}}$	35	–	17.8	35	–	39.57
σ	$\mathrm{S\,m^{-1}}$	215	–	0.18	215	–	0.33795
θ at 0 % SOC	–	0.0279	–	0.8490 (D)	0.0188 (D)	–	0.7025 (D)
θ at 100 % SOC	–	0.9014	–	0.2661	0.81	–	0.0038
Source		C; E of D_s from O			P; E of D_s from P12		

Note: k has units $\mathrm{A\,m^{-2}(m^3\,mol^{-1})^{1.5}}$, so that (9) gives i_0 in $\mathrm{A\,m^{-2}}$.

NMC811 cathode, from [4]:

$$U_p^{\mathrm{NMC}}(\theta) = -0.8090\,\theta + 4.4875 - 0.0428\tanh\big(18.5138(\theta-0.5542)\big) - 17.7326\tanh\big(15.7890(\theta-0.3117)\big) + 17.5842\tanh\big(15.9308(\theta-0.3120)\big). \quad (36)$$

$LiFePO_4$ cathode, as coded in PyBaMM's `Prada2013.py`:

$$U_p^{\mathrm{LFP}}(\theta) = 3.4077 - 0.020269\,\theta + 0.5\,e^{-150\,\theta} - 0.9\,e^{-30(1-\theta)}. \quad (37)$$

PyBaMM attributes this fit to [16]. In the arXiv text of [16], as read when assembling this set, the first exponential of the discharge fit is $e^{-200\theta}$; the PyBaMM form is used here so that the two codes can be compared directly. The difference matters only for $\theta \lesssim 0.03$ (the last few percent of charge).

5.2 Electrolyte properties

NMC cell ($LiPF_6$ in EC:EMC 3:7, Nyman et al. [5] as used in [4]), with $y = c_e/1000$ ($\mathrm{mol\,L^{-1}}$):

$$D_e = 8.794\times10^{-11}\,y^2 - 3.972\times10^{-10}\,y + 4.862\times10^{-10} \quad [\mathrm{m^2\,s^{-1}}], \quad (38)$$

$$\kappa = 0.1297\,y^3 - 2.51\,y^{1.5} + 3.329\,y \quad [\mathrm{S\,m^{-1}}]. \quad (39)$$

LFP cell (as coded in PyBaMM's `Prada2013.py`), with $z = c_e\times10^{-6}$:

$$D_e = 2\times10^{-10} \quad [\mathrm{m^2\,s^{-1}}], \quad (40)$$

$$\kappa = \big(4.1253\times10^{-4} + 5.007\,z - 4721.2\,z^2 + 1.5094\times10^{6}\,z^3 - 1.6018\times10^{8}\,z^4\big)\times10^{3} \quad [\mathrm{S\,m^{-1}}]. \quad (41)$$

For example $\kappa(1200) = 1.898\,\mathrm{S\,m^{-1}}$. The thermodynamic factor is 1 for both cells.

Table 2: Cell-level, electrolyte, thermal and side-reaction parameters.

Parameter	Unit	NMC cell	LFP cell
Plate area A	m^2	0.1027 (C)	0.18 (P)
Capacity Q, eq. (21)	Ah	5.0904 (D)	2.2999 (D)
Initial electrolyte conc. $c_{e,0}$	$\mathrm{mol\,m^{-3}}$	1000 (C)	1200 (P)
Transference number t_+	–	0.2594 (C)	0.36 (P)
Bruggeman exponent b (electrolyte)	–	1.5 (C)	1.5 (P)
Bruggeman exponent b_s (solid)	–	0 (C)	1.5 (P)
$D_e(c_e)$, $\kappa(c_e)$		eqs. (38)–(39) (C)	eqs. (40)–(41) (P)
E of D_e	$\mathrm{kJ\,mol^{-1}}$	17 (O)	26.6 (P12)
E of κ	$\mathrm{kJ\,mol^{-1}}$	17 (O)	11 (P12)
Heat capacity C_{th}	$\mathrm{J\,K^{-1}}$	68 (A)	77 (P12)
Cooling surface A_s	m^2	5.31×10^{-3} (C)	6.34×10^{-3} (P12)
Convection coefficient h	$\mathrm{W\,m^{-2}K^{-1}}$	25 (user input) (A)	25 (user input) (A)
$i_{0,\mathrm{sei}}$	$\mathrm{A\,m^{-2}}$	1.5×10^{-7} (O)	1.5×10^{-7} (O)
U_{sei}	V	0.4 (O)	0.4 (O)
α_{sei}	–	0.5 (O)	0.5 (O)
E of SEI rate	$\mathrm{kJ\,mol^{-1}}$	38 (O)	38 (O)
$i_{0,\mathrm{pl}}$	$\mathrm{A\,m^{-2}}$	0.01 (A)	0.01 (A)
α_{pl}	–	0.5 (A)	0.5 (A)
Derived quantities, for checking a re-implementation			
Anode active surface S_n	m^2	3.3597	2.1298
Cathode active surface S_p	m^2	2.967	323.14
Solid resistance R_s	mΩ	1.3645	1.0086
U_{ocv} at 0 / 10 / 100 % SOC	V	2.5457 / 3.3009 / 4.1929	2.0450 / 2.9816 / 3.5984
$\Phi_{p,\max} = U_p(\theta_{p,100})$	V	4.2849	3.6904

5.3 Remarks on the parameter sets

- **NMC capacity window.** The stoichiometry limits tabulated in [4] give different capacities for the two electrodes. Here the anode window defines the capacity and the cathode's 0 % stoichiometry is recomputed from (21) (0.8490 instead of 0.9084). The cathode stoichiometry at 100 % SOC (0.2661) was not re-checked against the paper; PyBaMM's initial condition corresponds to 0.2700.

- **NMC heat capacity.** $C_{\mathrm{th}} = 68\,\mathrm{J\,K^{-1}}$ assumes a 68 g cell at $1000\,\mathrm{J\,kg^{-1}K^{-1}}$. The layer properties in [4] give about $43\,\mathrm{J\,K^{-1}}$.

- **LFP capacity window.** PyBaMM's file gives the 100 % stoichiometries. The anode's 0 % value is set here so that the window holds the nominal 2.3 Ah; the cathode's follows from (21).

- **LFP values not from Prada et al.** The graphite open-circuit potential and both exchange-current laws are borrowed (see tag P). Reference [15] itself uses constant exchange-current densities (0.5 and $0.05\,\mathrm{A\,m^{-2}}$) and activation energies of 20 and $13\,\mathrm{kJ\,mol^{-1}}$ for charge transfer, lists three literature values (39, 43.8, $50.7\,\mathrm{kJ\,mol^{-1}}$) for the activation energy of D_s in $\mathrm{LiFePO_4}$ of which the first is used here, and its web text gives $D_{s,p} = 5.9 \times 10^{-20}\,\mathrm{m^2\,s^{-1}}$ where PyBaMM has 5.9×10^{-18}.

- **LFP physics.** $\mathrm{LiFePO_4}$ is phase-separating. Fickian diffusion in a sphere with a single-valued open-circuit curve ignores the two-phase front and the charge/discharge hysteresis.

Table 3: Discretisation, controller settings and default scenario.

Setting	NMC cell	LFP cell
Radial shells N_r per particle	20	
Electrolyte cells $N_n + N_s + N_p$	$10 + 3 + 10$	
Base time step Δt_0	2 s (4 s for sweeps)	
Electrolyte floor c_{fl}	$1\ \mathrm{mol\,m^{-3}}$	
Governor horizon H	3 steps	
Governor $c_{e,\min}$, $\theta_{n,\max}$	$100\ \mathrm{mol\,m^{-3}}$, 0.99	
Governor $\eta_{\min}$, $T_{\max}$ (defaults)	+5 mV, 45 °C	
Governor $I_{\max}$ (default)	3C	4C
CC-CV rate (default)	1C	2C
CC-CV voltage $V_{\max}$ (default)	4.20 V	3.60 V
Scenario (default)	$10\,\% \to 80\,\%$ SOC, $T_{\mathrm{amb}} = 25\,^{\circ}\mathrm{C}$, $h = 25\ \mathrm{W\,m^{-2}K^{-1}}$	

- **Side reactions.** The SEI constants are generic values from [6], not fitted to either cell; the plating constants are assumed. Capacity-loss figures are meaningful as comparisons between controllers, not as lifetime predictions.

6 Verification and benchmark results

6.1 Lithium conservation

With the side reactions switched off ($i_{0,\mathrm{sei}} = i_{0,\mathrm{pl}} = 0$), the script `tests/conservation.js` monitors

$$n_s = \sum_{j\in\{n,p\}} \varepsilon_{s,j} L_j A\, \bar{c}_s^{\,j}, \qquad n_e = A \sum_j \varepsilon_j \Delta x_j\, c_j, \qquad e_Q = \Big| \varepsilon_{s,n} L_n A\,(\bar{c}_s^{\,n} - \bar{c}_{s,0}^{\,n}) - \frac{1}{F}\sum_k I^k \Delta t^k \Big| \Big/ n_s \tag{42}$$

over 14 inputs per cell (constant current from 1C to 4C, discharge, random ±3C pulses with random step sizes, a sinusoid, rest, and closed-loop CC-CV and governor runs from −10 to 25 °C, including cases that deplete the electrolyte). The largest relative drifts are 5×10^{-14} for n_s, 7×10^{-13} for n_e and 2×10^{-14} for e_Q, i.e. floating-point round-off. Inventories at 10 % SOC: NMC $n_s = 0.282696$ mol, $n_e = 5.3677 \times 10^{-3}$ mol; LFP $n_s = 0.088318$ mol, $n_e = 1.24351 \times 10^{-2}$ mol.

6.2 Cross-check against PyBaMM

The scripts `tests/crosscheck_export.js` and `tests/crosscheck_pybamm.py` compare matched cases against PyBaMM 26.10 [13] (its SPMe and DFN models). For a like-for-like comparison the side reactions and the temperature dependences that this tool adds to transport are switched off.

- **NMC cell (Chen2020 set).** Voltage within 0.4–4 mV RMS of PyBaMM's SPMe up to 1.5C and 3–16 mV RMS of its DFN (PyBaMM's own SPMe is 3–13 mV from its DFN); anode surface stoichiometry within 0.003; temperature within 0.1–2.3 K RMS provided PyBaMM's "heat of mixing" option is enabled, consistent with (16). The plating overpotential (15) differs from the DFN by 3–13 mV RMS and reads up to about 13 mV *high* through the middle of a charge. Replaying the default governor current ($\eta_{\min} = 5$ mV) in the DFN gives a minimum of −8.0 mV; with $\eta_{\min} = 15$ mV the DFN minimum is +2.1 mV.
- **LFP cell (Prada2013 set, isothermal).** Voltage within 3–13 mV RMS of both PyBaMM models, dominated by a start-up transient in the anode surface concentration that 20 radial

points resolve coarsely in both codes (they agree within 2 mV when both use 80); below 1 mV after 2 min at 1C. The minimum of (15) agrees with the DFN within 0.5 mV up to 2C and in the governor replay (+5.0 against +5.2 mV).

6.3 Benchmark results

Table 4 gives the results of the default scenario for both cells, computed with the settings of Table 3. A re-implementation that follows Sections 2–4 should reproduce these closely; differences in the last digits can arise from the bisection path and from floating-point details.

Table 4: Benchmark: 10 % → 80 % SOC at $T_{\mathrm{amb}} = 25\,^{\circ}\mathrm{C}$, $h = 25\,\mathrm{W\,m^{-2}K^{-1}}$, $\Delta t_0 = 2\,\mathrm{s}$.

	NMC cell		LFP cell	
Quantity	CC-CV 1C	Governor	CC-CV 2C	Governor
Charge time [s]	2580	1662	1260	1080
Final SOC [–]	0.8004	0.8002	0.8000	0.8001
Q_{sei} [mAh]	0.3565	0.3558	0.2044	0.2179
Q_{pl} [mAh]	0.2907	0.0000	0.4989	0.0000
$\min \eta_{\mathrm{pl}}$ [mV]	−9.48	+5.00	−17.85	+5.00
Time with $\eta_{\mathrm{pl}} < 0$ [s]	276	0	462	0
Peak temperature [°C]	33.36	41.94	28.91	30.49
Peak voltage [V]	4.2000	4.2100	3.4352	3.4154
Peak current [C]	1.000	3.000	2.000	4.000
Governor: time each limit was active [s]				
none ($I = I_{\max}$)		12		54
plating overpotential		1602		1026
electrolyte depletion		48		0
Fastest CC-CV inside the governor's η_{pl} *and* T *limits* ($\Delta t_0 = 4\,\mathrm{s}$)				
rate [C] / time [s]	0.7117 / 3544		1.3445 / 1880	

6.4 Known limitations

- The SPMe assumes a uniform reaction current in each electrode. It loses validity when the electrolyte depletes at the anode (for the NMC cell, constant current of roughly 2–3C at 25 °C, or lower rates when cold).
- The single-particle estimate (15) of the plating overpotential cannot capture that particles next to the separator fill first. For the NMC cell this is a 10 mV-scale error (Section 6.2).
- The governor has perfect knowledge of the state. A practical implementation needs a state observer (for example [1]) and margins for model error.
- Aging is tallied but not fed back, so the model cannot be cycled as is.

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